

Single Python API

One interface for building, running, and analyzing molecular simulations

● Build

Prepare molecular systems

- Explicit / implicit solvent
- Ligand parameterization (GAFF2)
- Complex assembly
- Constant-pH-ready systems
- Coarse-grained (CALVADOS)



● Simulate

Run and control simulations

- Equilibration / production
- Checkpoint / restart
- Custom forces
- Constant-pH MD
- EVB free energy
- Umbrella sampling



● Analyze

Extract insights from data

- Conformational clustering
- Interaction-energy fingerprints
- Covariance-based PPI
- MM-PBSA (parallel)
- SASA
- Interface scoring (ipSAE / pDockQ)

Parsl execution layer

Scalable execution across environments

laptop · PBS / SLURM · cloud